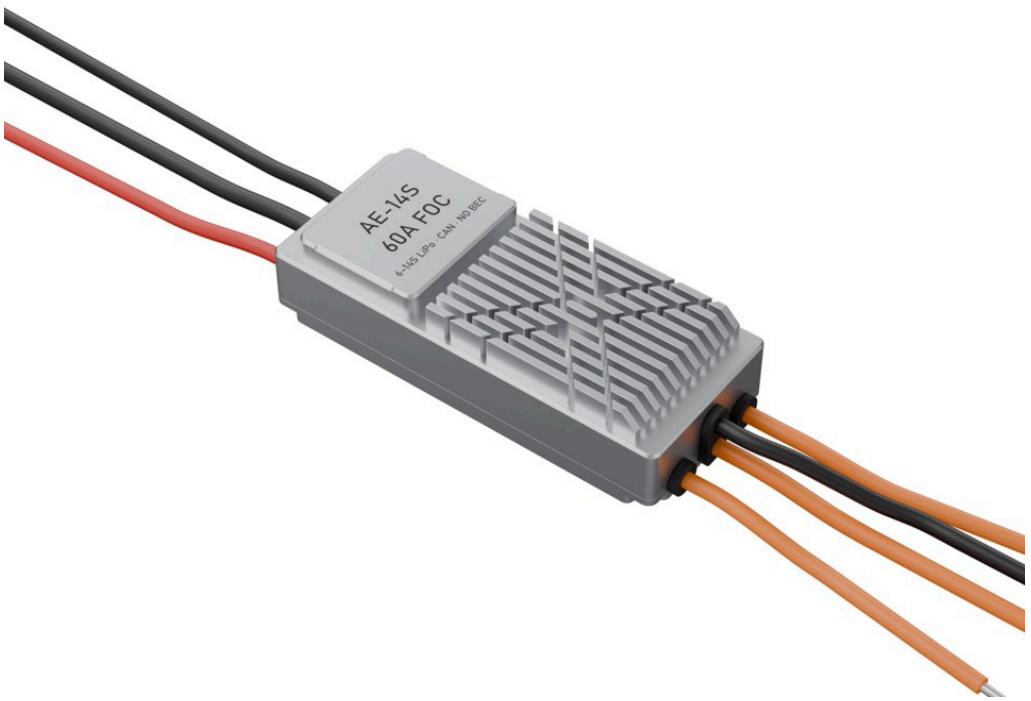
AE-14S 60A FOC

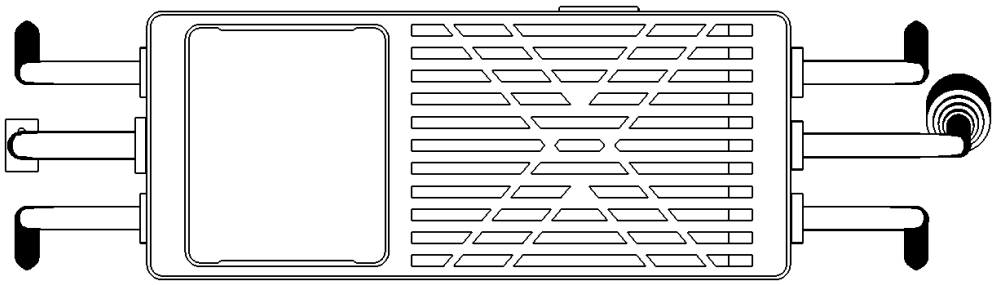
User Guide

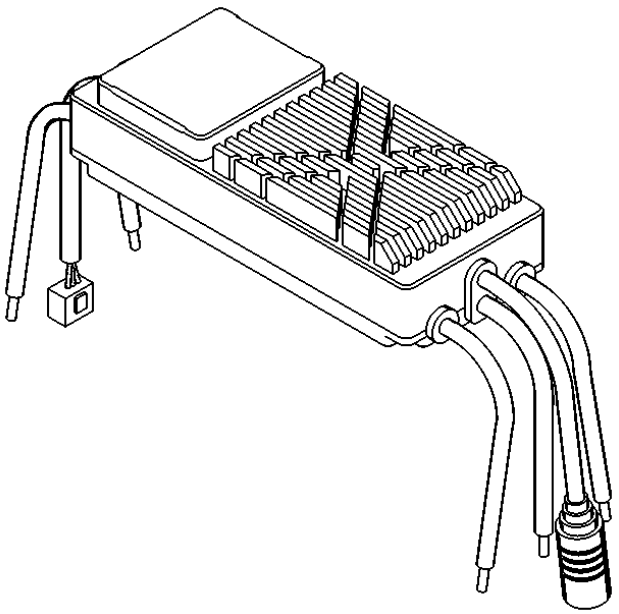


novaX

1. Parts List

· ESC





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1. Parts List

· Wires & Cables

Wire	Type	Signals
Battery leads	14 AWG · battery end · 2 wires	BAT+ · BAT−
CAN cable	battery end · JST-GH 4P plug · 3 wires	CAN_H · CAN_L · GND
PWM input (option)	battery end · selected with can_pwm = 1 · 2 wires	PWM · PWM−
Motor leads	orange · 16 AWG · motor end · 3 wires	MOT_A · MOT_B · MOT_C
Encoder cable	motor end · round 6-pin plug (Ø 8.3 mm) · SPI · 6 wires	3V3 · GND · CS · CLK · MOSI · MISO

2. Specifications

· Hardware Specifications (1)

MCU	TI TMS320F28027F + ST STM32F103
Control	Field-oriented control with the motor's magnetic encoder
Voltage Range	6–14S LiPo (18–60 V)
Constant Current	30 A
Burst Current	60 A (10 s)
Interface	DroneCAN (CAN 2.0 · 1 Mbit/s · isolated) on a JST-GH 4P cable throttle, telemetry, settings and firmware updates
Telemetry	DroneCAN ESC status voltage, current, rpm, temperature and fault flags
Prop Parking	Holds the propeller at a stored angle at 0 % throttle
Protection	Current limited every switching cycle · heat lowers the limit, never below 50 % · short-circuit retry · voltage and throttle-loss flags
Motors	novaX AM-6212 / AM-6215 with encoder (28 poles), measured on the aircraft with auto_cal

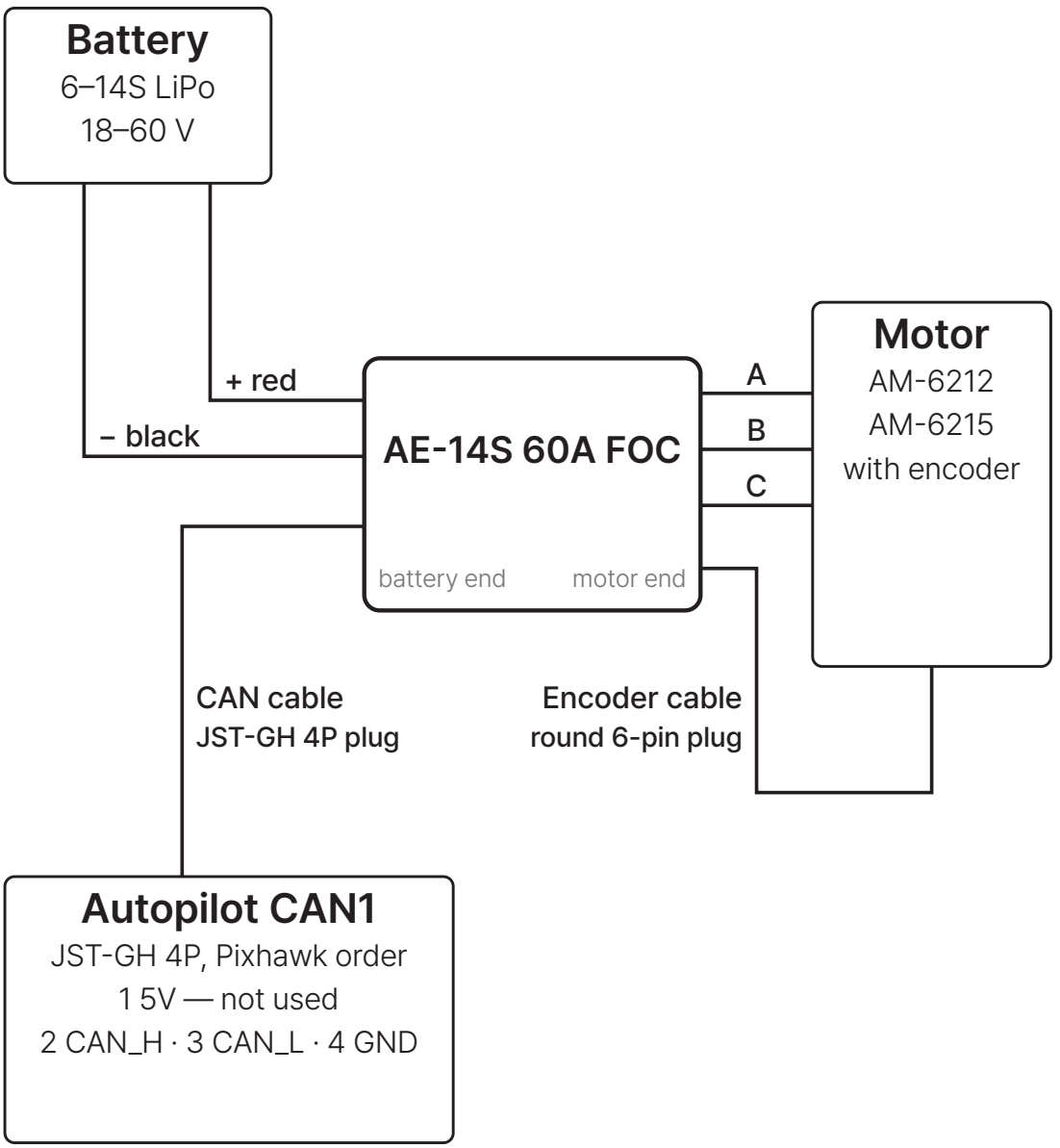
2. Specifications

· Hardware Specifications (2)

Wires	Battery 14 AWG · motor 16 AWG · CAN cable with JST-GH 4P plug · encoder cable with round 6-pin plug
Size	84 × 35 × 20 mm
Firmware	novaX FOC App V1.62 · DroneCAN 1.14 with CAN bootloader V1.2, both updated over DroneCAN

3. Wiring

· Connection Diagram

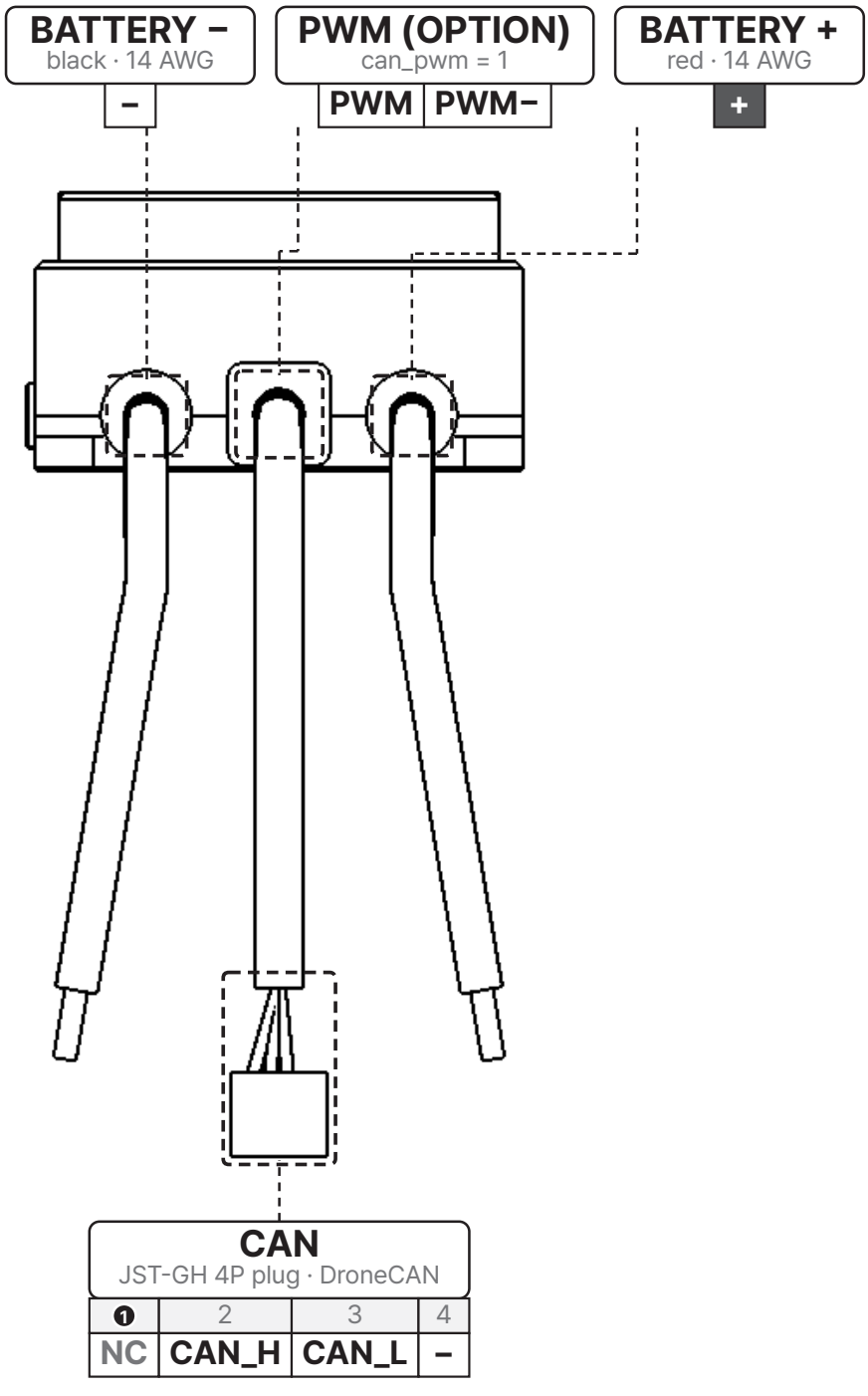


Motor leads: any order — then run auto_cal and set the direction with dir. CAN carries the throttle, telemetry, settings and firmware updates.

3. Wiring

· Battery End

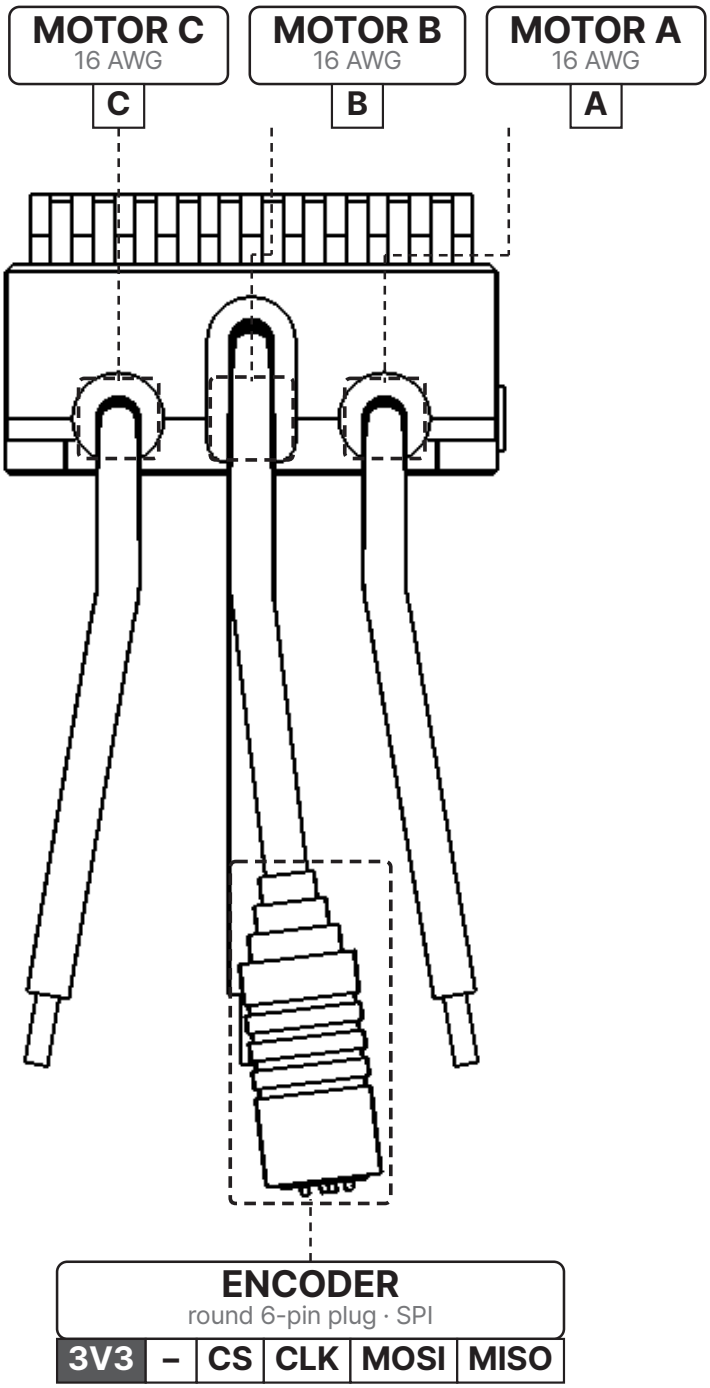
❶ = pin 1 **+** = battery **3V3** = supply **–** = GND **CS** = signal · seen from outside the case



3. Wiring

· Motor End

❶ = pin 1 **+** = battery **3V3** = supply **–** = GND **CS** = signal · seen from outside the case



4. Setup

· Step by Step (1)

Set the autopilot first (parameters below), then give every ESC its number. ESC parameters are DroneCAN parameters: in Mission Planner open SETUP → Optional Hardware → DroneCAN/UAVCAN, choose MAVlink-CAN1 and press Start. Each ESC appears as AE14S_60A_V1.62 (the motor app version) — Menu → Parameters, change the value, Write. The ESC refuses a change while its motor turns or the autopilot is armed (the old value comes back), so disarm first.

1. Wiring

Battery leads to the power distribution (red +, black –). The three motor leads to the motor in any order. Plug the encoder cable's round 6-pin plug into the motor's encoder cable and the CAN cable's JST-GH 4P plug into the autopilot CAN1 port (its pin 1, 5 V, is not used: the ESC powers its isolated CAN side itself).

2. Autopilot

Set the ArduPilot parameters in the table and reboot. CAN_D1_UC_ESC_BM marks the outputs that are sent to DroneCAN ESCs (15 = outputs 1–4); CAN_D1_UC_ESC_OF skips that many outputs, so the first marked output becomes ESC 0 (motors on outputs 5–8: BM 240, OF 4).

3. ESC number

Give each ESC its esc_index: 0 for the first motor output, 1 for the next, and so on. Number the ESCs one at a time with only that ESC connected, or spin each with the motor test and match the rpm shown.

4. Setup

· Step by Step (2)

4. Motor measurement

Propeller off, motor cold, autopilot disarmed: write auto_cal = 1. The motor turns slowly for about 20 s while the ESC measures resistance, inductance and flux twice and calibrates the encoder. motor_meas = 1 means the values were saved; any other number tells why not (see the user guide).

5. Direction

Spin the motor with the motor test at 7–10 %. If it turns the wrong way, change dir (0 ↔ 1). Do not swap motor leads after auto_cal: the encoder calibration follows the lead order (after a swap, run auto_cal again).

6. Prop parking (optional)

Fit the propeller, turn it by hand to the parking angle and write prop_align = 1. At 0 % throttle the ESC holds the propeller there. Write 1 again to store a new angle, 0 to switch it off.

4. Setup

· Autopilot Parameters

Set these on the autopilot and reboot it.

Parameter	Value	Description
CAN_P1_DRIVER	1	Enable the CAN1 port with driver 1 (reboot after the change)
CAN_P1_BITRATE	1000000	1 Mbit/s (default)
CAN_D1_PROTOCOL	1	DroneCAN
CAN_D1_UC_ESC_BM	15	Outputs sent to DroneCAN ESCs: 15 = outputs 1–4 (motors 1–4). 0 = no CAN throttle at all
CAN_D1_UC_ESC_OF	0	Outputs skipped before ESC 0. Motors on outputs 5–8: CAN_D1_UC_ESC_BM 240 and CAN_D1_UC_ESC_OF 4

4. Setup

· ESC Parameters (1)

In Mission Planner: SETUP → Optional Hardware → DroneCAN/UAVCAN → MAVlink-CAN1 → Start, then Menu → Parameters on the ESC (AE14S_60A_V1.62). Change the value and Write.

Parameter	Value	Description
esc_index	0 ... 19	Position in the CAN throttle command: 0 = first marked output, 1 = the next ... Every ESC needs its own number (default 0)
node_id	0 / 1 ... 125	0 = node number from the autopilot (dynamic allocation, default); 1–125 = fixed number
dir	0 / 1	Spin direction. Change it here, never by swapping motor leads
prop_align	0 / 1	1 = store the present propeller angle and hold it at 0 % throttle; 1 again = new angle; 0 = off. Motor stopped, after a passed auto_cal
auto_cal	1 / 2	1 = measure the motor and calibrate the encoder (about 20 s, slow turn, propeller off, disarmed, cold motor). 2 = forget the measured values (defaults from the next power-up). Reads 0
can_pwm	0 / 1	Throttle source: 0 = CAN (default), 1 = PWM input (option, see the pinout)

4. Setup

· ESC Parameters (2)

Parameter	Value	Description
sound_off	0 / 1	1 = no start-up tones (do-mi-sol) at power-up; 0 = tones on (default). CAN chip 1.14 or newer
motor_rs_mohm	read only	Phase resistance measured by auto_cal (mΩ, 0 = none)
motor_ls_uh	read only	Inductance measured by auto_cal (µH, 0 = none)
motor_flux_vphz	read only	Flux measured by auto_cal (V/Hz, 0 = none)
motor_meas	read only	Result of the last auto_cal: 1 = saved; 11–23 = not saved, the reason is listed in the user guide
motor_pole_pairs	read only	Pole pairs found by auto_cal (14 for AM-6212 / AM-6215, 0 = none)

5. auto_cal Result

· motor_meas

After auto_cal the ESC shows the result in motor_meas.

Code	Meaning
0	Not measured since this power-up
1	Measured values saved
11	Failed: resistance
12	Failed: inductance
13	Failed: flux
14	Values out of range — not saved
15	The motor was turning — not measured
16	The rotor did not follow — not saved (check the propeller and for binding)
17	Stopped: the aircraft was armed
18	After an in-air restart — power-cycle, then try again
19	The motor may be hot from running — let it cool, then try again
20	The ESC is hot — let it cool, then try again
21	Autopilot disarm not confirmed — not started
22	The two passes did not agree — the previous values are kept
23	The motor chip did not answer

6. Firmware

· Update over DroneCAN

Firmware is updated over DroneCAN through the autopilot — the case stays closed. Latest: motor chip app V1.62 and CAN chip 1.14 with CAN bootloader V1.2 (2026-11-05). Disarm the autopilot and stop the motor (the ESC refuses an update otherwise), then send the .bin with a DroneCAN firmware updater — Mission Planner (SETUP → Optional Hardware → DroneCAN/UAVCAN → MAVlink-CAN1 → Start, then the ESC's Menu → Update Firmware) or the novaX PC Throttle tool. One file type per chip; the ESC reads the header and sends it to the right chip. The node name shows the motor app version (AE14S_60A_V1.62). A blank or unbootable motor chip is reflashed by novaX over JTAG.

An update takes about 45 seconds. Keep the battery connected until the ESC restarts and its node name shows the new version.

7. Safety

· Before You Fly

- Check the battery polarity before connecting — a reversed battery destroys the ESC at once.
- Stay within 6–14S LiPo (18–60 V).
- Remove the propeller for auto_cal and whenever you work near the motor; fit it only to set the parking angle.
- Do not swap motor leads after auto_cal — change the direction with dir.
- Keep the battery leads short: long leads raise the voltage spikes the ESC has to take.
- Mount the ESC in the airflow. Heat lowers its current limit (down to 50 %), it does not stop the motor.
- Settings and firmware updates are refused while the motor turns or the autopilot is armed.

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